Critical Replication Paper

Food insecurity, affected by socio-demographic factors, is associated with water insecurity?

A cross-sectional analysis of sub-Saharan Africa

By Davod Ahmadi (2019)

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Declaration:
I, Xolani Charles Ncala, declare that this work is my own. I have not plagiarised anyone's work. All sources have been cited using standard referencing conventions.

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1 Introduction

Over the past two decades reproducible research has become an important element in scientific research, as it allows scholars to be able verify findings, detect possible errors and confidently advance certain aspect of prior work, while fostering a culture of transparency and cumulative knowledge. Since the American Political Science Association (APSA) published the symposium debate on replication policies in political science, multiple journals within the discipline have introduced data sharing or replication policy, encouraged scholars to produce replicable work, while others made it a requirement condition before publishing scholarly work (King, 2006). In "Publication, Publication", Gary King (2006) expresses that starting with a replication of a study puts your work right where the field is most active, where if you manage to refine or rethink even one part of the original study in a way that is meaningful and evident, your contribution can advance knowledge on the subject, therefore making it "publishable". This paper then aims to make a contribution towards development literature focused on Water Insecurity and Food Insecurity by critically replicating statistical analyses from Davod Ahmadi (2019) "Food insecurity, affected by socio-demographic factors, is associated with water insecurity? A cross-sectional analysis of sub-Saharan Africa", then improving on the author's statistical analyses, and thereafter reporting on the new insights or results discovered.

Ahmadi's main goal with the study is to examine the extent of the relationship between water insecurity and food insecurity across 32 Sub-Saharan African (SSA) countries, as well as to explore how the socio-economic and demographic characteristics impact the prevalence and severity of food insecurity. The point of departure for his study comes from recognising that across the world, inadequate access to water and food remains among the most critical challenges of resource scarcity that is impacting millions of individuals, "a challenge that is especially severe in low-income and developing countries" (Ahmadi, 2019). According to Ahmadi (2019) Sub-saharan Africa faces the greatest impact from water scarcity, with "a larger proportion of its population lacking reliable access to clean water sources and improved water services". The challenges with not ensuring consistent availability of safe water is that it also has a major effect on fulfilling the society's capacity to produce food process food for nourishment, and that makes sense why the Food and Agriculture Organisation (FAO) found that that "more than 300-million people in SSA are severely food insecure" (Ahmadi, 2019; FAO, 1996). With these realisations Ahmadi (2019) study primarily centers on regions facing major challenges related to both water and food access aiming to investigate how reliable water supply influences individual's ability to be food secure for their own nutritional needs.

The article conceptually frames *food insecurity* "as the lack of enough food to maintain a healthy and productive life" and *water insecurity* "as the absence of consistent access to adequate water for a healthy living" (Ahmadi, 2019). The data used used for the study was obtained from the Afrobarometer Household Survey Round 6 conducted in 2016 which only included only data from the 32 African nations of focus (Ahmadi, 2019). Analyses were conducted using IBM Statistical Package for the Social Sciences (SPSS) which included variable descriptive statistics, bivariate associations, and implementing two Multinomial Logistic Models (MLMs), which generate final odds ratios to assess whether a one-unit increase in water insecurity levels is linked to a corresponding change in the likelihood of the food insecurity when controlling for socio-economic and demographic variables [Area of residence, Age-group, Education, etc.] (Ahmadi, 2019). The findings indicate "a significant and positive association between water insecurity and food insecurity" (Ahmadi, 2019) therefore demonstrating that individual respondents who had reliable access to water were over 34 times ($OR = 34.08$ more likely to avoid experiencing any form of food insecurity. The socio-demographic factors also showed to significantly influence patterns of food insecurity, as individuals who had adequate financial resources were more prone to experience food stability, while some of those who lacked formal schooling faced substantially reduced odds of being food secure $OR = 0.141$, similarly living in rural areas was more associated with less food insecurity $OR = 0.457$ (Ahmadi, 2019). The study's findings notably highlight that within the SSA Afrobarometer sample, the experience of going without water is strongly associated with the experience of going without food, this is the same for some demographic factors included as control variables like income availability, education, and area of residence.

There are a few the major concerns to note with regards to the process of the replication study. The first relates to the condition that there is no replication code and dataset provided by the publishing journal and author to actually guide the replication process in the decisions he made with the statistical analyses. While the study's coding decisions were not available, it was possible to replicate the author's key findings using independently constructed models and variable definitions aligned with the published methodology. The second is that, instead of the IBM SPSS the data analysis procedures were carried out utilising RStudio (version R.4.4.2), a software platform that incorporates an interpreter tailored for the R programming language, which is specially developed for performing statistical computations. With regards to the improvements proposed in relation to the replicated results they focus on the author's decision to use mulitinomial logistic regression model when the output variable and the main dependent variable have an ordinal nature. The usual standardard practice with multinomial logistic regression is that it is applied for multi categorical outcomes whereas ordinal variables are modeled through Ordinal Logistic Regression known as the proportional odds model (Fox & Weisberg, 2018; Liang et al., 2020). With these considerations, the paper will first explore the conceptual underpinnings of food insecurity and water insecurity, by explaining some of the factors that cause them, and advancing some of Ahmadi (2019) ideas on how they manifest onto society. The second part of the paper is made of the replication process that provides details to the precise steps taken to replicate all of the author's statistical analyses displayed on tables, graphs, and regression summaries from the original article. The third segment of this replication paper is the critical improvement section where following the identification of a potential area for improvement, we proceed to implement and analyse it. In this particular instance it means it will discuss the results and the broader implications of applying ordinal logistic regression as an improved approach. The fourth and last part of the paper forms the conclusion phase where we harmonise the results of the main paper by Ahmadi (2019) with the improved regression analyses.

2 Water and Food Insecurity in Context: Causes, Consequences, and Evolving Metrics

Ahmadi (2019) explains water insecurity "as the absence of consistent access to adequate water for a healthy living", this is the widely accepted definition among developmental studies including Frongillo (2023) who argues that it is not only a condition of having too little water for basic use but "it is about the quality, accessibility and reliability of water sources". Its causes are multifaceted, including inadequate infrastructural development, unequal distribution of water resources, conflicts over water access, climates variability and pollution. Water insecurity has profound effects on individuals, as it impacts their health through increased risks of water borne diseases, compromises their sanitation and hygiene which ultimately disrupts their livelihoods especially for communities reliant on water for agriculture (Muller et al., 2009). According to Frongillo (2023)recent development literature has made advances in moving from the physical and infrastructural indicators of water insecurity which often masked the lived realities of individuals standardized scales and surveys such as the Household Water Insecurity Experiences (HWISE) that center the personal, emotional and behavioral dimensions of water insecurity. These measurement metrics and tools have made improvements to capture how people cope and adapt to water related challenges across the diverse cultural and geographic contexts. The Afrobarometer Household Survey is an example of these improvements where a sample of respondents for research can now directly answer questions about their experiences of having water or going without water.

With food insecurity as the lack of access to food for a healthy life, its main causes go beyond mere food availability and therefore includes factors such inadequate income, rising food prices, poor agricultural growth and socio-economic inequalities. Frongillo (2023) study found that some of the consequences of inadequate food access severely affect children, who end up having stunted growth due to their weakened immune systems, so this means that the impact is not only physical but also emotional as it puts pressure on families who do not know where their next meal would come from. Similar to water insecurity, food insecurity has also moved away from the focus of agricultural output on food products to experienced based measures from surveys or proxy questions that measure how many times individuals go without food.

In most times water and food insecurity do not exist in isolation from each other as they feed into one another therefore making those faced with them experience difficult living conditions. For instance when a household struggles to get enough clean water it does not only affect their thirst but it affects how food is grown cooked and stored. On another lens the when there is food shortage or scarcity people often turn to options that possibly demand more water, like staple foods that need long time to cook. According to Walsh and Van Rooyen (2015) the intersection going without water or going without food is largely shaped by deeper social and economic realities, for instance if your income is low and you live far from reliable city infrastructure you are more likely to face both kinds of insecurity. So essentially studies like Ahmadi (2019) articles are important because they are not just about statistics, but about reporting the exact struggles faced by individuals on a daily basis.

3 Replication Process

The data for the study come from Afrobarometer Round 6 data obtained from a multiple public opinions surveys done across 38 African countries, but at the time Ahmadi (2019) conducted the study he only chose to focus on data available for the 32 SSA countries, from over a total of [n = 49 137] individual respondents. Afrobarometer has become widely known for the measurement of multiple aspects about African nation's health, politics, economics, governance and social struggles, it primarily checks on how the selected sample of individuals feel about these aspects. According to Ahmadi (2019) the survey researchers make sure that they select a sample with traits or characteristics that will reflect the real population, and interviews are done with "randomly selected sample of $n = 1200$ or $n = 2400$ people in each country". This method is widely known as the "stratified, multi-stage, area probability sampling" simply meaning that they carefully selected people by first dividing the country into key groups, then narrowing down step by step from regions to households to individuals, making sure that every respondent has a fair and equal chance of being selected (Ahmadi, 2019). The data is made publicly available on the Afrobarometer website where you can download their code books, interview questions, and data sets available in *.csv* files or SPSS.

From the survey, the author measures food insecurity using question "Over the past year, how often, if ever, have you or anyone in your family: Gone without enough clean water for home use?". The values of this question are: "0= Never, 1= Less than once a year, 2= At least once a year, 3= At least every six months, 4= At least every three months, and 5= At least once a month". These are the same response values with the main dependent variable water insecurity which asks "Over the past year, how often, if ever, have you or anyone in your family: Gone without enough clean water for home use?", and the socio-economic control variable income insecurity which asks "Over the past year, how often, if ever, have you or anyone in your family: Gone without a cash income?" (Ahmadi, 2019). There are also other control variables including Asset ownership which was created based on whether whether a respondent owns items like a radio, TV, vehicle, and mobile phone ranging from 0 to 4. This where some of the confusion with Ahmadi's methods starts to drill in when looking at the first three mentioned variables they the questions are right but the values of the responses are incorrect, for instance the Afrobarometer Round 6 merged code book shows the response values for food insecurity, water insecurity and income availability to be: 0=Never, 1=Just once or twice, 2=Several times, 3=Many times, 4=Always, 9=Don't know, 98=Refused to answer, -1=Missing", so meaning that the correct order of the response is coded from 0 to 4 with 9, 98 and -1 being non response values that are recoded and treated as NAs during replication process. When importing the data on the R studio it reveals the same idea that Ahmadi (2019) might have been mistaken about the value specifications of these variables, this is also evident in the results of his statistical analyses where the variable categories are not exactly named as he proposes on his methodology section.

The replication process follows a specific order of how Ahmadi produces some of the statistics contained in his tables, graphs, and models (see,section 5), in an attempt to see whether is the study generally replicable with only using methodological guidelines without access to the original code. We start of off with a replication of Figure 2 which contains the sample sizes of the studied countries and their Gross Domestic Product measured by Power Purchasing Parity (GDP.PPP), followed by Figure 3 which contains the characteristics of sample of SSA countries, and then following with a display of Figure 4 which is the bivariate graph of those who have never experienced food insecurity and water insecurity from from the 32 countries. The replication then moves to replicating the bivariate cross-tabs analyses which Ahmadi (2019) conducted to explore the associations between dependent, independent and control variables on Figure 5and then lastly we will run the unadjusted multinomial model on Figure 6 and the unadjusted multinomial model on Figure 7.

3.1 Replicating the Sample size and GDP.PPP Table

Ahmadi's (2019) data analyses procedures begin with a display of the the sample size of the studied countries and the four GDP PPP groups that they are divided into (see, Figure 2). However, the table offers no further interpretation or theoretical framing beyond its descriptive function. It primarily appears to be only intended to show the distribution of the sample of respondents and to signal the SSA countries' relative economic standing of the sample of respondents and to signal the SSA countries' relative economic standing or purchasing power, without elaborating on how these economic tiers relate to the article's core outcomes. In the original Figure 2 from the study the sample sizes are understandable to be calculated from the Afrobarometer data set, with regards to the GDP.PPP groupings there is a slight indication that they are based on

World Bank data from 2018, but Ahmadi (2019) has not provided the original figures on top of the unclear significance of the GDP.PPP groupings. This is one of reasons why when replicating I have decided to additionally use the World Bank data to import the real figures of the GDP.PPP per country so that we can have a solid background for these GDP.PPP groupings and communicate the implications of the findings, as an economic measure among then countries.

Table 1: Sample size of studied countries in sub-Saharan Africa (n = 49,137)

GDP PPP Group	Country	GDP PPP (2018, $)	Sample Size
GDP PPP ($4,000 and more)	Mauritius	23068.24	1200
	Botswana	15785.82	1200
	Gabon	15432.04	1198
	South Africa	13347.36	2390
	Namibia	9853.74	1200
	Swaziland	8536.99	1200
	Cape Verde	7916.42	1200
	Ghana	5536.50	2400
	Nigeria	5083.17	2400
	Cote d'Ivoire	4945.63	1199
	Kenya	4411.56	2397
	Sudan	4199.25	1200
	Cameroon	4010.76	1182
GDP PPP ($2,000 to $3,999)	Sao Tome and Principe	3941.03	1196
	Zambia	3442.27	1199
	Senegal	3380.22	1200
	Benin	2965.31	1200
	Guinea	2844.46	1200
	Tanzania	2727.88	2386
	Mali	2684.41	1200
	Sierra Leone	2640.48	1191
	Lesotho	2632.29	1200
	Zimbabwe	2614.41	2400
	Uganda	2312.06	2400
	Burkina Faso	2073.32	1200
	Togo	2067.09	1200
GDP PPP ($1,500 to $1,999)	Liberia	1799.73	1199
	Madagascar	1546.60	1200
GDP PPP (Less than $1,500)	Malawi	1363.79	2400
	Mozambique	1325.59	2400
	Niger	1276.00	1200
	Burundi	753.46	1200

Reference: Micro data analysis of Afrobarometer's data in 2016; World Bank, 2018.

The replication process of the Figure 2 results on the R studio statistical softaware began with the loading of the Afrobarometer Round 6 data as .csv file and the GDP PPP data for the 32 countries as a .csv file obtained from the World Bank economic indicators dataset. We then standardised the country names and fultered the data set to match the conutry list on to which the ISO3 codes (standardized three-letter abbreviations used to represent countries) were attached through the countrycode package to enable merging with GDP data. We then selected the relevant survey variables and renamed them for clarity including the insecurity measures and demographic factors. It is from this step then that we can now have the basis of the GDP.PPP group divisions backed by data into the four income brackets that reflect Ahmadi's economic tier groupings (Isbell, 2017).

In the final step we summarise the sample sizes by country and GDP.PPP group in a sorted descending order as you can see on the final output in Table 1 which starts with the nation with the highest GDP.PPP being Mauritius with a $23 068 GDP.PPP and forms part of the ($4000 and more) gdp group. The main finding with the of the replicated table is that it provides advanced statistical findings from Ahmadi (2019) result on the same table, mainly we can see that by manually classifying GDP.PPP groups without the exact data points on which grouping is based, Ahmadi falls victim to the misclassification of several countries including Swaziland which was originally placed under the ($2000 - $3999) but its actual 2018 GDP.PPP grouping should be in the ($4000 or more) group, Sao Tome and Principe is incorrectly placed under the ($1500 or less) group but its data point on GDP.PPP shows that it belongs in the ($2000 - $3999).

With the replication of this table there were new insights gained that could have a significant contribution and meaning in relation to the study's objective of determining factors that affect food insecurity. For starters Table 1 which I have successfully replicated, reveals a wide economic gradient across the sub-Saharan African nations, with GDP.PPP ranging from ($23,068) to Burundi at ($753), demonstrating the vast disparities in purchasing power these nations have. Most of the countries fall below the ($4,000) mark and the ($2,000) mark, which most likely means that most of the countries are faced with economic vulnerability of having less purchasing power. The sample sizes are largely uniform at around 1,200 respondents per country, with key exceptions to countries like South Africa, Nigeria, and Malawi as their samples go above 2000 respondents, likely to enhance a nationally representative sample. The economic indicator accompanying the sampling patterns, are significant for this study because usually a nation's economic capacity like GDP.PPP often correlates with either a heightened or reduced exposure to food and water insecurity, for instance households in poor economies are less equipped to withstand economic disruptions and maintain consistent access to infrastructure and service. In this sense the GDP.PPP variable then can possibly be a valuable contextual moderator - (Ahamadi does not do this) - to help researchers to test whether economic conditions amplify or buffer the impact of water insecurity on food insecurity outcomes.

3.2 Replicating The Characteristics of Sample Table

Ahmadi's (2019) results section begins with the presentation of his descriptive finding of the data analyses which are based on the figures displayed in Figure 3 "Characteristics of sample of SSA countries ($n = 49,137$)". The amount of respondents from the SSA sample who reported to have had secure access

Table 2: Characteristics of respondents (n = 49,137)

Variable	**Category**	**Frequency**	**Percent**
Area of residence	Urban	19825	40.3
	Rural	29312	59.7
Sex	Male	24401	49.7
	Female	24736	50.3
Age group	18–25	11944	24.4
	26–35	14876	30.4
	36–45	9885	20.2
	46–55	6023	12.3
	56–65	3708	7.6
	66 and over	2438	5.0
Education	No formal education	9223	18.8
	Primary	14403	29.4
	Secondary	18323	37.4
	Post-secondary	7057	14.4
Employment	Unemployed	30045	61.4
	Employed part-time	5898	12.1
	Employed full-time	12966	26.5
Income availability	Never	11176	22.8
	Just once or twice	5823	11.9
	Several times	11485	23.5
	Many times	13028	26.6
	Always	7416	15.2
Electricity access	No	18662	38.0
	Yes	30459	62.0
Water insecurity	Never	25929	52.9
	Just once or twice	5523	11.3
	Several times	7872	16.1
	Many times	5592	11.4
	Always	4127	8.4
Food insecurity	Never	24300	49.6
	Just once or twice	7311	14.9
	Several times	9470	19.3
	Many times	5700	11.6
	Always	2177	4.4
Asset ownership	Owns all 4	6654	13.5
	Owns 3	13896	28.3
	Owns 2	12949	26.4
	Owns 1	10073	20.5
	Owns none	5565	11.3

to water round up to 53% of the total sample while, respondents who reported never going without food amounted to about 52% of the total sample, this means that close to 47% of the sample does experience some level of water insecurity, and approximately 48% of the sample experiences some level of food insecurity. Ahmadi (2019) constructed Figure 3 to also present the frequencies of the control variables, and their different categories, for instance we can now see the Afrobarometer gender balance showing as 50.3% of the sample are female and the remaining 49.7% are male, and some concerning results on the table show that in all of the sample there is 61% unemployment rate, and approximately only 15% of respondents had some level of post-secondary education. These are alarming in the context of this study, because high unemployment rate points to the presence of systematic economic challenges, and low post-secondary education rate is concerning because it affects the progress of development and job opportunities available within a country, these two elements are negative contributors to both food security and water security.

With regards to the replication process of this table this is where a lot of the coding decisions were becoming tricky to replicate mainly because the authors coding decisions are not something that are typically included in the article's methodology process, these are some of the elements that are usually specified on the replication code book or script. This process was successful after we were able to systematically recode and standardised key variables, by cleaning invalid responses, collapsing categories and also grouping numeric variables as factors according to Ahmadi's framework. The categorical divisions of Figure 3 also guided the recoding process because we knew the divisions that needed to be created from the available dataset. For instance when recoding the Gender variable, it was first cleaned to remove the invalid or non-response codes and recoded on R studio as 0= Male, and 1= Female. What was also noted during this process is that for this table Ahmadi (2019) used "pairwise deletion", meaning that the non-response values coded "NA" are only being removed for the specific variable that is being analysed while the rest of the row on the dataset stays intact.

With regards to the specific recoding of the main dependent variable, the main independent variable and the income availability variable are not resembling the same divisions that were highlighted by Ahmadi (2019) methodological stages where he had said that the questions for those variables have response values ranging from "0= Never, 1= Less than once a year, 2= At least once a year, 3= At least every six months, 4= At least every three months, to 5= At least once a month". Table 2 and the Afrobarometer Round 6 merged codebook expose this contradiction because the real response values to those questions are "0=Never, 1=Just once or twice, 2=Several times, 3=Many times, to 4=Always", a nature that is ordinal because it reflects an increasing frequency although its not assumed that moving from one level to the has equal spacing (Isbell, 2017). With that being one of the mistakes that Ahmadi (2019) makes, the other confusion originates from the recoding of the Area of Residence variable, when performing a few checks on the variable we found that it is not naturally dichotomous, because it has more than two response factors, being "1= Urban, 2= Rural, 3= ?, 460= ?". The latter two levels have unknown labels that the Afrobarometer Round 6 codebook (Isbell, 2017) does not account for, making up a combined respondent frequency of $n = 600$ (3= 512; 460 = 88). So with the weighting of the Area of residence variable for the original study close to 600 responses had to be treated as non responses and dropped from analysis, it is therefore concerning because its relatively a large number that could had some significance to the analysis.The solution that came up for this confusion, since there are some Afrobarometer rounds conducted after the 6th one, the successive round 7 data code book (Han, 2020) revealed that the real values for this variable are "1=Urban, 2=Rural, 3=Semi-Urban, 460=Peri-Urban", so to keep the dichotomous nature of the variable in the replication we recoded everything with urban as "0 = Urban" and the rural as "1 = Rural". We can see that the only difference to Ahmadi's original Figure 3, is that the replicated Table 2 recoded urban category in the Area of residence variable is actually 40.3% not 39.6% and the rural is 59.7% not 60.4%.

There is a control variable that is included in the original Figure 3 which is the electricity variable, a dichotomous variable that indicates whether the respondents have state installed electricity connection to their residence or household and we can see that 62% of the total sample said yes and the remaining 38% said no. From Ahmadi (2019) point of view there is no confirmation as to why he included that variable in Figure 3, because it does not get examined in any of the consequent statistical analyses he conducted after this table including the bivariate association crosstabs and the multinomial models. Going forward, the decision has been to get this variable is included in the remainder of the replication of Ahmadi (2019) results and also added on in the critical analyses section to find out what effect does having electricity or going without it have on respondents' experience of food insecurity. With all of the above mentioned problems considered, explained, and addressed, the table is a successful replication.

3.3 Replicating the Food secure and Water secure Graph

This part of the replication process was concerned with replicating the first and only graphical visualisation included on the original paper by Ahmadi (2019) (see,Figure 4), showing the percentages of respondents who reported "never gone" without water and those that reported "never gone" without food across the 32 SSA nations, as a form of variable proportions test. On the main paper Ahmadi (2019) reported that, "people living in countries with access to water were more likely to be food secure", and that the "probability of water and food security are shown to be significantly high within countries with high GDP per capita like Mauritius, South Africa, and Namibia".

The replication of these results was fairly simple on the R studio software, the major change made to the graph as you can see from Figure 1, is to make visible the water secure percentages across the nations just to enhance the

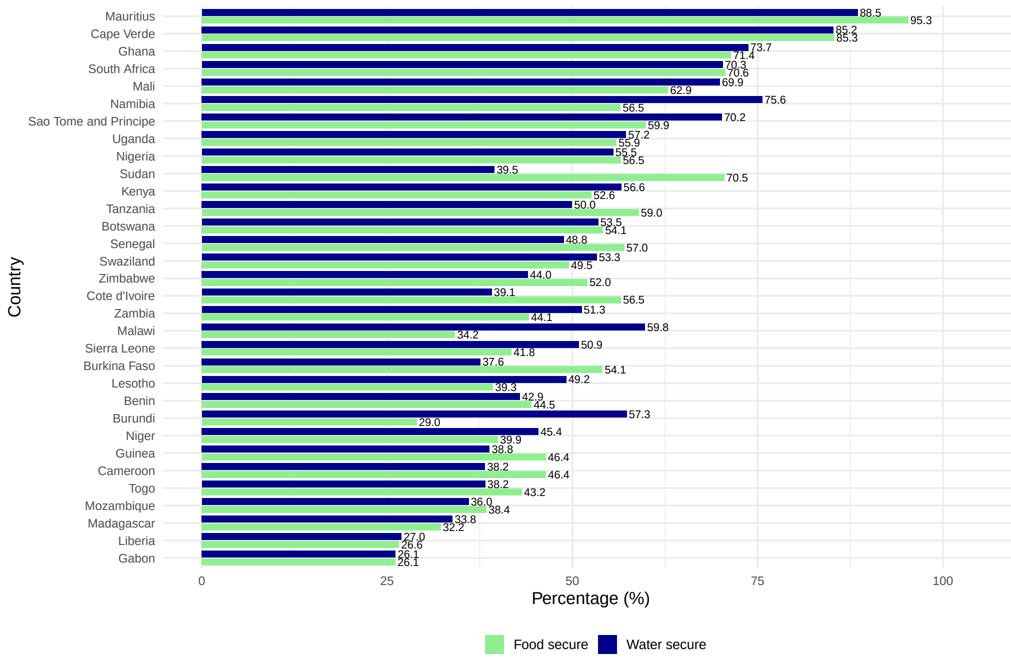

Figure 1: Water and food security in SSA (Afrobarometer Round 6).

interpret-ability of the graph. These adjustments help us see the exact figures that in many countries water security is lower than food security especially in countries like Sudan where 70% of the sample was food secure and only about 40% of the sample was water secure, and Ivory Coast with 56% food secure and only 39% water secure. When viewing the graph from the GDP.PPP perspective from Table 1 we can see that the lowest GDP.PPP countries like Malawi, Mozambique, and Niger struggle with low food and water security percentages most of which go below 45% of the total sample, and the observed anomalies include Gabon which has a higher GDP.PPP at $15,532, but has the lowest percentages of both water and food security at 26.1% and 26.1% respectfully. These numbers from the sample suggest that when it comes to Gabon its economic wealth alone does not guarantee that most respondents are food and water secure.

3.4 Replication of the bivariate analyses

The original study Figure 5 displays the results of the bivariate analyses between food insecurity, the predictor variables and controlling factors, conducted by Ahmadi (2019) to explore existing associations between them. The bone of contention when it comes to production of the statistics represented on the table the author specifies that he ran cross-tabs analyses, which computed the Chi-Square significance estimates (used to determine whether two categorical variables are associated or independent of each other) on the Sig.level column, and the Strength statistic which is not specified on the methods and analyses as to where it originates. The analyses of the table also only speak of the significance statistic and less about what the strength statistic is, for instance Ahmadi reported that; "All socio-demographic factors were found significantly related to food insecurity; however, income availability (0.581; $p < 0.0001$), asset ownership (-0.334; $p < 0.0001$), employment (-0.233; $p < 0.0001$), education (-0.283; $p < 0.0001$), and area of residence (0.137; $p < 0.0001$) were observed to be strongly associated with food insecurity". With understanding the ordinal nature of the dependent variable and main independent variable, a research of statistics used to do a bivariate analyses on ordinal variables revealed that the statistic might be Crammer's V which has values between 0 and 1 and the Spearman's rank correlation which takes on values between -1 and $+1$ (Anderson et al., 2025). This gave the idea that ultimately the Strength column presented the spearman rank correlation coefficient which is "a measure of the monotonic relationship between two variables when data in the form of ranks are used", at $+1$ it indicates strong positive association and at -1 it indicates a strong negative association (Anderson et al., 2025).

Table 3: Independent Spearman's rho correlations between food insecurity and socio-economic factors in SSA countries (n = 49,137).

Variable	Strength (Spearman's ρ)	Sig. level
Water insecurity	0.400	< 0.001
Gender	0.025	< 0.001
Age group	0.063	< 0.001
Area of residence	0.137	< 0.001
Education	-0.227	< 0.001
Employment	-0.152	< 0.001
Income availability	0.500	< 0.001
Electricity	-0.200	< 0.001
Asset ownership	-0.285	< 0.001

The replication process with this particular table in Figure 5 was not as simple because the estimated figures in Table 3 do not exactly match the estimates presented by Ahmadi (2019), but the direction of the relationships across both tables is consistent for all the variables although the magnitudes from Ahmadi's Figure 5 are larger. For instance for water insecurty the spearman's correlation is at $(0.400, p < 001)$ instead of $(0.485, p < 0.001)$, this is not a major difference and might only be as a result using two different statistical soft wares, but they both confirm the same result that there is a positively strong and significant association between water insecurity and food insecurity. Some of the control variables which have a negative influence on the rise of food insecurity include asset ownership $(-0.285, p < 0.001)$, education $(-0.227, p < 0.001)$, and the newly added variable electricity $(-200, p < 0.001)$. So far we can see that water insecurity has an obvious connection with food insecurity when analysed in isolation. The next section deals with checking the effect that water insecurity has on food insecurity when controlling for the socio-demographic variables through the replication of the article's two multinomial regression models.

3.5 Replication of the Multinomial Regression Models

Table 4: Unadjusted multinomial logistic regression analyses of Food Insecurity by socio-economic factors in SSA countries (n = 49,137)

		Food Insecurity Levels			
		Odds Ratios (95% CI)			
Variable	**Category**	**Never**	**Just once or twice**	**Several times**	**Many times (Ref: Always)**
Water insecurity	Never	39.176 (33.407–45.941)	13 (10.918–15.478)	7.902 (6.703–9.315)	4.609 (3.894–5.456)
	Just Once Or Twice	16.575 (12.831–21.41)	27.834 (21.365–36.261)	10.508 (8.1–13.631)	3.61 (2.745–4.747)
	Several Times	11.241 (9.161–13.793)	13.796 (11.11–17.131)	13.464 (10.956–16.547)	5.423 (4.384–6.709)
	Many Times	3.493 (2.949–4.138)	3.626 (3.005–4.376)	4.022 (3.388–4.775)	4.82 (4.06–5.722)
Ref: Always					
Gender	Female	0.828 (0.74–0.926)	0.891 (0.791–1.003)	0.88 (0.783–0.989)	0.956 (0.846–1.079)
Ref:Male					
Age group	18–25	2.322 (1.838–2.934)	2.263 (1.756–2.916)	1.591 (1.246–2.032)	1.346 (1.043–1.738)
	26–35	1.818 (1.456–2.272)	1.94 (1.522–2.472)	1.462 (1.158–1.845)	1.263 (0.99–1.611)
	36–45	1.484 (1.18–1.868)	1.581 (1.231–2.031)	1.339 (1.053–1.703)	1.219 (0.948–1.566)
	46–55	1.625 (1.263–2.091)	1.597 (1.215–2.101)	1.378 (1.059–1.794)	1.47 (1.118–1.932)
	56–65	1.407 (1.07–1.85)	1.449 (1.076–1.951)	1.194 (0.896–1.59)	1.45 (1.079–1.95)
Ref: 66 - Over					
Area of residence	Rural	0.457 (0.403–0.518)	0.63 (0.552–0.719)	0.756 (0.663–0.861)	0.984 (0.858–1.127)
Ref: Urban					
Employment	Unemployed	0.563 (0.463–0.684)	0.498 (0.407–0.61)	0.705 (0.576–0.862)	0.799 (0.648–0.985)
	Employed Full-Time	1.859 (1.466–2.357)	1.137 (0.889–1.453)	1.108 (0.867–1.416)	1.049 (0.812–1.354)
Ref: Part-time					
Income availability	Just Once Or Twice	0.527 (0.35–0.794)	4.214 (2.772–6.406)	1.748 (1.137–2.686)	1.414 (0.889–2.251)
	Several Times	0.284 (0.206–0.391)	2.003 (1.438–2.789)	3.71 (2.653–5.188)	2.008 (1.397–2.886)
	Many Times	0.11 (0.082–0.147)	0.916 (0.675–1.244)	2.486 (1.824–3.387)	4.39 (3.149–6.12)
	Always	0.012 (0.009–0.016)	0.121 (0.091–0.162)	0.361 (0.27–0.483)	0.821 (0.599–1.123)
Ref: Never					
Asset ownership	Owns None	0.065 (0.05–0.086)	0.208 (0.157–0.276)	0.398 (0.301–0.527)	0.873 (0.651–1.171)
	Owns 1	0.168 (0.129–0.22)	0.425 (0.322–0.562)	0.729 (0.552–0.963)	1.17 (0.873–1.569)
	Owns 2	0.27 (0.207–0.354)	0.677 (0.512–0.895)	0.927 (0.701–1.225)	1.113 (0.828–1.496)
	Owns 3	0.49 (0.372–0.645)	0.807 (0.606–1.074)	0.894 (0.671–1.192)	0.921 (0.68–1.249)
Ref: Owns 4					
Education	No Formal Education	0.141 (0.111–0.179)	0.25 (0.194–0.321)	0.51 (0.397–0.654)	0.975 (0.749–1.269)
	Primary	0.238 (0.187–0.302)	0.515 (0.401–0.659)	0.722 (0.563–0.925)	1.113 (0.856–1.447)
	Secondary	0.536 (0.42–0.684)	0.846 (0.657–1.091)	0.984 (0.763–1.27)	1.083 (0.827–1.418)
Ref: Post-secondary					
Electricity access	No Electricity	0.345 (0.308–0.386)	0.515 (0.457–0.58)	0.751 (0.668–0.844)	1.012 (0.896–1.143)
Ref: Has Electricity					

Reference: Micro data analysis of Afrobarometer's data in 2016

This section of the replication focusses on the replication of Ahmadi (2019) two multinomial logistic regression models (see, Figure 6, Figure 7). According to Liang et al. (2020) the multinomial logistic regression model is a "statistical tool that is used to analyse specific set of data where the outcome variable has more than two categories", while the usual binary logistic regression model is limited to predicting a variable with only two possible outcomes, the "multinomial model extends to accommodate several distinct, unordered outcome variables categories at once" (Fox & Weisberg, 2018; Liang et al., 2020). It helps to quantify how each of predictors influence the likelihood of falling into a specific outcome group (Food insecurity - Never) compared to the designated reference category (Food insecurity - Always) and then reports the effects as log-odds coefficient which show how "moving up one unit on the predictor shifts the odds of being in a given category relative to the reference" (Liang et al., 2020). These can be converted to odds ratios for easier interpretation as Ahmadi does in both the original unadjusted model and the adjusted model. On R studio statistical software the multinomial model is computed through the multinom() function contained within the nnet package by B. Ripley et al. (2016), it is mainly by virtue of this package being available that enabled the replication of a multinomial model on R.

Figure 6 contains the results of Ahmadi's unadjusted multinomial regression models between the predictor variables and the outcome variable food insecurity with its response levels shown at the top of the table. The important key-word with this output is the name "unadjusted", which refers to when the regression models the effect of a sinlge independent variable on a multi-category outcome "without acoounting for potential confounders or co-variates"according to Liang et al. (2020). So for instance Ahmadi's unadjusted mulinomial regression basically estimates how water insecurity affects the odds of being in each food insecurity category while completely ignoring the effects of other factors, this is usually a process done to explore the raw bivariate associations between the variables, and its limitation is the tendency to overstate the effects of a variable due to omitted variable bias. Ahmadi (2019) initial report stated that individuals with sufficient access to clean water for household use were significantly more likely to report no food insecurity, with an odds ratio of 39.17 (95% CI: 33.40–45.93), the income availability variable emerged as the strongest predictor of food insecurity with an odds ration of 83.95(95% CI: 63.89-11.03), both are significant predictors of all levels of food insecurity. The replicated Table 4 shows the exact same estimates that were originally obtained by Ahmadi, it only differs with the advancement of including the Electricity access variable, which shows to be a significant predictor of food insecurity for the first three levels ("Never," "Just once or twice," and "Several times"), meaning that repondents without electricity are significantly less likely to be food secure with an odds ratio of 0.345 (95% CI: 0.308–0.386). Ahmadi(2019) does not explicitly specify that he uses the unadjusted model to try an eliminate the variables that possibly can create a multicollinearity problem in the adjusted model (see, Figure 7), hence he excludes education, income availability and asset ownership variables, so that they do not distort the main effect of water insecurity.

When it comes to Figure 7, the adjusted multinomial logistic regression differs with one element from the unadjusted, which is that it includes multiple forms of predictors at the same time, so that we can compute the independent effect of each predictor after "adjusting" or controlling for others. It simply adjusts for the confounding influence of other predictors. For instance when

Table 5: Adjusted multinomial logistic regression analyses between water insecurity and food insecurity by controlling for gender, age, area of residence, employment, and electricity in sub-Saharan Africa (n = 49,137)

		Food Insecurity Levels			
		Odds Ratios (95% CI)			
Variable	Category	Never	Just once or twice	Several times	Many times (Ref: Always)
Water Insecurity	Never	32.136 (27.294–37.838)	11.679 (9.771–13.960)	7.718 (6.523–9.131)	4.731 (3.981–5.621)
	Just Once Or Twice	14.119 (10.871–18.337)	25.359 (19.367–33.204)	10.394 (7.974–13.548)	3.688 (2.789–4.876)
	Several Times	10.056 (8.146–12.413)	13.049 (10.451–16.294)	13.633 (11.035–16.843)	5.741 (4.616–7.140)
	Many Times	3.223 (2.707–3.838)	3.527 (2.910–4.275)	4.032 (3.382–4.806)	5.088 (4.267–6.066)
Ref: Always					
Gender	Female	0.840 (0.745–0.947)	0.889 (0.784–1.009)	0.875 (0.773–0.989)	0.966 (0.851–1.096)
Age	18–25	2.212 (1.720–2.846)	2.170 (1.658–2.839)	1.575 (1.217–2.039)	1.357 (1.039–1.771)
	26–35	1.556 (1.222–1.983)	1.705 (1.316–2.209)	1.387 (1.083–1.777)	1.226 (0.950–1.583)
	36–45	1.226 (0.955–1.573)	1.354 (1.037–1.768)	1.260 (0.977–1.626)	1.175 (0.903–1.528)
	46–55	1.441 (1.097–1.894)	1.480 (1.106–1.980)	1.387 (1.049–1.832)	1.503 (1.129–2.000)
	56–65	1.258 (0.938–1.688)	1.353 (0.989–1.851)	1.166 (0.863–1.575)	1.435 (1.056–1.952)
Ref: 66 - Over					
Urban/Rural	Rural	0.883 (0.755–1.032)	1.015 (0.863–1.194)	0.963 (0.820–1.130)	1.057 (0.895–1.249)
Ref: Urban					
Employment	Unemployed	0.617 (0.503–0.757)	0.556 (0.450–0.686)	0.767 (0.623–0.945)	0.804 (0.648–0.997)
	Employed Full-Time	1.592 (1.244–2.038)	1.098 (0.852–1.415)	1.060 (0.823–1.364)	0.993 (0.765–1.289)
Ref: Part-time					
Electricity Access	No Electricity	0.634 (0.549–0.732)	0.789 (0.679–0.916)	1.052 (0.909–1.218)	1.242 (1.068–1.445)
Ref: Has Electricity					

Model Statistics: -2 Log Likelihood (Full Model): 110,967.115; Likelihood Ratio Chi-square: 14,516.807, df = 56, p = 0; Pseudo R-squared (CoxSnell / Nagelkerke / McFadden): 0.222 / 0.240 / 0.098; AIC = 111,087.115; BIC = 111,613.986.

comparing Ahmadi (2019) unadjusted model (see, Figure 6) to the adjusted one (see,Figure 7), we can see the effect of all levels the water insecurity status has on all food insecurity levels while holding gender, age, geographical area, and employment constant, with slightly reduced odds ratios on all levels of food insecurity compared to the baseline always (see, Figure 7). In practical terms it means that even after adjusting for controls to factor in the effect of other socio-demographic water insecurity still remains as a significant predictor.

The adjusted multinomial model replicated on Table 5 was a successful replication with no complications. The added electricity access variables made did not have a major impact on the model estimates produced the original adjusted model estimates. The replication result shows that water insecurity odds ratios are still highly consistent, therefore indicating a monotonic decline in food insecurity risks as water insecurity decreases, with odds ratio of $OR = 3.223 (95\% CI : 2.707 - 3.383)$ for the (many times) water insecure in the food (never) category (see, Table 5. In both the original (see, Figure 7) and replicated versions of the adjusted multinomial model (see,Table 5 being female slightly reduces the odds of being food secure with

(95% CI: 0.84-0.85). A surprising result also shows that in Ahmadi (2019) adjusted model without the electricity access variable (see, Figure 7) the (rural) category had a significant reduction in the odds ratio of food security $OR = 0.697(95\%CI : 0.610 - 0.796)$, but for the Table 5 results electricity access takes off some of the effect the rural category had and leaves it only with an $OR = 0.883(95\%CI : 0.755 - 1.032)$ which is statistically insignificant. For the employment variable, categories (unemployed) and (employed full-time) are interpreted relative to part-time workers, for instance the result show that unemployed respondents from the sample consistently have lower odds of being food secure that part-time workers on all food insecurity levels with $OR = 0.38$ (never food insecure) on the authors original table and $OR = 0.62$ (never food insecure) on the replicated version with electricity access. With regards to model fit statistics included below the original adjusted model(see, Figure 7) and the replicated one (see, Table 5, it reveals that both models show strong statistical significance which is an indication that the predictors used in each one does a solid job of explaining food insecurity patterns. Table 5 of course includes a few more variables that Ahmadi's model therefore giving it a slightly better fit overall, but less difference when it comes to both model's explanatory power as they account for about 21-24% of the variation in food insecurity based Cox & Snell and Nagerlkerke measures, and for the Mcfadden R^2 the variation explained is around 9-10%. Essentially while the replicated adjusted model (see,Table 5) edges ahead in terms of some fitting statistics both models seem to be performing at a similar level that confirms that water insecurity and socio-demographic factors are meaningful predictors of the outcome.

4 The Ordinal Regression Model Output

The ordinal regression model also referred to as the proportional odds model, is a particular statistical method used to analyse data with an ordinal dependent variable, which means that "its categories have a natural order but the intervals between the categories are not necessarily equal" (Liang et al., 2020). According to Liang et al. (2020) in R studio statistical software the ordinal regression output includes t-statistics for each predictor, which can be used to derive p-values of the coefficients through the Wald test significance. The model presents an accurate estimate of the log odds of falling at or below a given level of the outcome compared to the higher level, across all thresholds, as it operates under the proportional odds assumption, which implies that the influence of each predictor remains consistent regardless of where the cut-off points are drawn on the ordinal outcome scale (Fox & Weisberg, 2018; Liang et al., 2020). The statistical package for the proportional odds model is the MASS in the R studio statistical software built by B. D. Ripley (2002) as part of the "*Modern Appliead Statistics With S*" textbook, it contain function "polr()" used to formulate ordinal regression models. The Sub-Saharan African sample of the Afrobarometer Round 6 surveys examined in Ahmadi

(2019) study, has the food insecurity response variables in a naturally ordered nature, this means that its categorical structure needs to be preserved without the assumption of equal spacing between the levels, and it is for this reason that "the proportional odds model offers a statistically appropriate and interpretable framework for analysis" (Liang et al., 2020).

From the results produced during the Replication procedure, there are multiple reasons that necessitate modelling an ordinal regression model, based on the conceptual frame work of food insecurity which represents a continuum of deprivation. Essentially, the multinomial logistic regression used in Ahmadi (2019) study, that treats the outcome categories of the response variable as nominal and unordered, does not have the OLR capability to exploit the inherent ordering of categories for analysing their cumulative probability. In incorporating this approach the aim is not only to follow the conceptual framework of increasing severity, but the added advantage is that it reduces the number of parameters needed for modelling, to increase its statistical efficiency and predictive power (Fox & Weisberg, 2018). In addition, there are also similar ordinal variables like our independent variable of interest water insecurity and several controls like income insecurity, asset ownership, education, which are also in an ordinal nature, so using the OLM aligns analytically with the likert-type scaling of these variables to maintain the consistency across the measurement of both dependent and independent variables.

It is clear that before running the the ordinal regression some variables needed to be recoded from factors with labels to likert item ordinal variables. For instance with the main predictor water insecurity soci-economic control variable income insecurity in **??** were initially categories from (never) to (always), in this model it has been cleaned and recoded from $0-4$ to a likert scale range of $1-5$ moving from never to Always. Ahmadi's analyses used the age variable as factored groups when, which is completely abandoned on this analyses as, age is taken as the exact numerical values of the respondent's age which fits an adult age range of 18 to 105, and the asset ownership variables was also recoded into a numeric scale of 0-4 which is a cumulative score of essential household assets thereby helping to control for the respondents socio-economic context. In **??**, the construction of the Table 1 included the the values of the Gross Domes Product Power Purchase Parity in dollars($) and it was grouped into 5 income brackets as factors with labels, this is so we can factor in the effects of being in different GDP.PPP economic tiers. In the instance that it is a factor this means that its estimates reflect the estimated effect of being in one economic group compared to the reference which is the ($\$4,000$ or more) category.

The results of the ordinal logistic regression in Table 7 strengthens the empirical case that the predictor water insecurity is not merely correlated to food insecurity, but it is structurally embedded in its patterns and how it affects individuals across the Sub-Saharan Africa region. Its odds ratio reads as $OR = 1.437(95\%CI : 1.417 - 1.458)$, at a highly statistically significant rate, meaning that each unit increase in water insecurity from (never)

Table 6: Summary of Variable Recoding Procedures

Variable	Original Coding / Values	Recoding Procedure and New Scale
Water Insecurity	0 = Never, 1 = Just once or twice, 2 = Several times, 3 = Many times, 4 = Always, 9 = Don't know, 98 = Refused, -1 = Missing	Replaced invalid codes (9, 98, -1) with `NA`; added +1 to shift valid range to **1–5** (1 = Never → 5 = Always).
Income Insecurity	0 = Never, 1 = Just once or twice, 2 = Several times, 3 = Many times, 4 = Always, 9 = Don't know, 98 = Refused, -1 = Missing	Same treatment as Water Insecurity — invalid codes replaced with `NA`; shifted to **1–5** scale (1 = Never → 5 = Always).
Education	0 = No formal schooling, 1 = Informal schooling (Koranic), 2 = Some primary, 3 = Primary completed, 4 = Some secondary, 5 = Secondary completed, 6 = Post-secondary diploma/college, 7 = Some university, 8 = University completed, 9 = Post-graduate, 99 = Don't know, 98 = Refused, -1 = Missing	Removed invalid codes (99, 98, -1); added +1 to shift valid range to **1–10** (1 = No formal schooling → 10 = Post-graduate).
Asset Ownership	"Owns none", "Owns 1", "Owns 2", "Owns 3", "Owns all 4"	Recoded to numeric scale **0–4** (0 = Owns none → 4 = Owns all 4).
Age	Continuous values; –1, 998, 999 = Invalid/Missing	Invalid values replaced with `NA`; retained valid range **18–105** as numeric variable.
GDP_PPP_Group	Factor GDP.PPP Groups(USD)	Categorized into five ordered groups: **1 = $1,000 and less**, **2 = $1,000–$1,999**, **3 = $2,000–$2,999**, **4 = $3,000–$3,999**, **5 = $4,000 and more**,(ref).

to (always) raises the odds of experiencing high food insecurity about 44%. Unlike the complex multinomial results, water insecurity in Table 7 shows a strong positive monotonic relationship, where the more frequently Sub-Saharan African households go without water the more they are likely to experience food deprivation. Similarly confirmed by Ahmadi's (2019) multinomial models in, Table 7 shows that income insecurity is the highest predictor in the model, as those who frequently go without income have an odds ratio of $1.994 (95\% CI : (1.960 - 2.028)$, meaning that they are twice as likely to be food

Table 7: Ordinal logistic regression (POLR) results for determinants of food insecurity in Sub-Saharan Africa (n = 49,137)

Variable	Category	Coefficient	OR (95% CI)	Interpretation
Water Insecurity	Water insecurity	0.363	1.437 (1.417–1.458)	Higher food insecurity risk
Gender	Female (vs Male)	-0.007	0.993 (0.956–1.031)	No significant effect
Area of Residence	Rural (vs Urban)	-0.109	0.896 (0.856–0.939)	Slightly lower odds
Age	Continuous (years)	0.005	1.005 (1.003–1.006)	Small positive association
Employment	Unemployed (vs Full-time)	0.095	1.100 (1.050–1.152)	Increased risk
	Part-time (vs Full-time)	0.137	1.146 (1.074–1.223)	Increased risk
Income Insecurity	Income insecurity	0.690	1.994 (1.960–2.028)	Strongly increased risk
Education	Education level	-0.061	0.941 (0.931–0.951)	Protective factor
Asset Ownership	Asset ownership	-0.225	0.799 (0.784–0.814)	Protective factor
Electricity Access	No electricity (vs Has electricity)	-0.002	0.998 (0.950–1.048)	No significant effect
GDP (PPP group)	$1,000 or less (vs Ref: $4,000 or more)	0.251	1.286 (1.149–1.438)	Higher odds
	$1,000–$1,999	0.391	1.479 (1.398–1.564)	Higher odds
	$2,000–$2,999	-0.367	0.693 (0.660–0.728)	Lower odds
	$3,000–$3,999	-0.181	0.834 (0.774–0.900)	Lower odds

Model statistics: AIC = 104,148.72; BIC = 104,306.84; logLik = -52056.36 N = 48,273; df = 18; Pseudo R^2 (McFadden / Cox–Snell / Nagelkerke) = 0.170 / 0.358 / 0.386.

insecure, which therefore reinforces the idea that financial stability is central to food access. When it comes to the other socio-demographic variables Table 7 reveals that higher education, greater asset ownership and being a rural resident are all statistically significant protective factors against food insecurity, and they are separately associated with reduced odds that reflect better access to resources or probably subsistence food production. The model reveals that for the respondents who do not have full time employment the odds of food insecurity increase, while the GDP.PPP variables shows some mixed effects as countries with a GDP (PPP) of $1,000 or less have about 28.6% higher odds of experiencing food insecurity compared to those in countries with a GDP of $4,000 or more, but the middle economic tiers have some relative protection their effects are all statically significant. This shows that extremely low national income levels are strongly associated with a greater risk of food insecurity. Lastly the ordinal model suggests that gender and electricity access have no statistically significant effect on food insecurity, which means that when controlling for all the other factors these variables do not independently influence risk.

At the bottom of Table 5 and Table 7 there statistics of fit for the models, which are used to determine which model better fits the data and its complexity. At first instance we can look at the AIC statistic which penalizes complexity lightly for better predictive accuracy and the BIC which penalizes model complexity more heavily for some parsimony, Table 7 has a much lower AIC and BIC which means it fits the data more efficiently and has improved parsimony. Both the McFadden and Nagelkerke pseudo R^2 values moving from Table 5 to Table 7, show some improvement as the Mcfadden almost doubles from $0.098- > 0.170$, and the R^2 rises from $0.240- > 0.386$, this is indicative of the greater explanatory power in the Table 7 ordinal regression model. Essentially by respecting the ordered nature of food insecurity responses, the model captures the cumulative probability across variable categories and reduces parameter redundancy and that is why the proportional-odds structure aligns better with the data's conceptual ordering than the multinomial model used by Ahmadi (2019).

5 Conclusion

With this replication study we set out to critically reproduce and refine Davod Ahmadi's (2019) report which, provides analyses of the association between water insecurity and food insecurity across the 32 SSA nations that were contained in the Afrobarometer Round 6 Survey dataset. The dual aim to see if it is possible to replicate a study such as Ahmadi's (2019), by basically replicating all of his main statistical analyses and findings including some discriptive, bivariate and multinomial regression findings, so that we can be able to implement some statistical advancements or solutions by building on his existing body of work. From the start of the section 3 the replications process up until the its end, the replication successfully recovered Ahmadi's (2019) core results that water insecurity is strongly linked to higher food insecurity with large odds rations that persist even after adjusting for the demographic and employment factors. Some of the highlights in terms of problems experienced during replication, include methodological and transparency issues, in particular the unavailability of replication code, inconsistencies between reported variablle coding and the Afrobarometer codebook, the ambiguous treatment of missing data specifically the Area of residence variable, and the exclusion of socio-demographic variable "electricity access" without computing it into some model or primary bivariate analyses. Crucially, Ahmadi's application of multinomial logistic regression treated the ordered outcome as a set of unrelated categories, therefore leasing to an inflated number of parameters that fail to capture the inherent cumulative structure of the data.

In order to address most of these limitations section 4 introduced the proportional-odds model/ the ordinal logistic model, that better aligns with the conceptual and statistical nature of the food insecurity variables. This model delivered three key improvements (see, Table 7). On first note it the ordinal model offered statistical parsimony and interpret-ability, with only fewer parameters and a single cumulative odds interpretation, for instance a one unit increase in water insecurity increased the odds of high food insecurity by 44% with $OR = 1.437$. On second note Table 7 exhibited enhanced model performance, showing reduced AIC/BIC and increased pseudo R^2 values (McFadden, Cox–Snell, Nagelkerke) in contrast to Ahmadi's multinomial method. The third note is that the ordinal regression's simplicity ensured conceptual consistency by treating the outcome and key predictors as ordinal, and reflecting the severity continuum inherent in food insecurity. Overall, while the replication confirms Ahmadi's substantive claim, it also exposes the reproducibility gaps of the study and offers a more elegant,efficient,and theoretically grounded modelling strategy that strengthens the original conclusions and sets a clearer path for future research.

References

Ahmadi, D. (2019). Food insecurity, affected by socio-demographic factors, is associated with water insecurity? a cross-sectional analysis of sub-saharan africa. *International Journal of Development and Sustainability*, *8*(9), 619–632.

Anderson, D. R., Sweeney, D. J., & Williams, T. A. (2025). Statistics [Encyclopedia Britannica, published 7 Nov. 2025. Accessed 12 Nov. 2025].

FAO. (1996). Rome declaration on world food security and world food summit plan of action [Accessed on 27 April 2025]. file://E:/FAO%20Document%20Repository_files/show_cdr_head.htm

Fox, J., & Weisberg, S. (2018). *An R companion to applied regression* (3rd). Sage Publications.

Frongillo, E. A. (2023). Intersection of food insecurity and water insecurity. *The Journal of Nutrition*, *153*(4), 922–923.

Han, K. (2020, December). *Data codebook for round 7 afrobarometer survey* (tech. rep.) (Prepared in collaboration with University of Cape Town (UCT), Centre for Social Science Research, and CDD-Ghana). Michigan State University. East Lansing, Michigan, USA.

Isbell, T. A. (2017). *Data codebook for round 6 afrobarometer survey* (tech. rep.) (Prepared in collaboration with CDD-Ghana and Michigan State University). University of Cape Town, Centre for Social Science Research. Rondebosch, South Africa.

King, G. (2006). Publication, publication. *PS: Political Science & Politics*, *39*(1), 119–125.

Liang, J., Bi, G., & Zhan, C. (2020). Multinomial and ordinal logistic regression analyses with multi-categorical variables using r. *Annals of Translational Medicine*, *8*(16), 982.

Muller, M., Schreiner, B., Smith, L., van Koppen, B., Sally, H., Aliber, M., & Pietersen, K. (2009). *Water security in south africa* (Working Paper Series No. 12). Development Planning Division. Midrand, South Africa. https://www.dbsa.org/sites/default/files/media/documents/2021-09/DPD%20Working%20Paper%2012.%20Water%20security%20in%20South%20Africa.pdf

Ripley, B. D. (2002). *Modern applied statistics with s*. Springer.

Ripley, B., Venables, W., & Ripley, M. B. (2016). *Package 'nnet'* [R package version 7.3-12]. https://CRAN.R-project.org/package=nnet

Walsh, C. M., & Van Rooyen, F. C. (2015). Household food security and hunger in rural and urban communities in the free state province, south africa. *Ecology of Food and Nutrition*, *54*(2), 118–137.

Appendix

Table 1. Sample size of studied countries in SSA (n= 49,137)

GPD PPP (4,000 and more $)	Cape Verde	1200
	Mauritius	1200
	South Africa	2390
	Botswana	1200
	Namibia	1200
	Gabon	1198
	Zambia	1199
	Ghana	2400
	Nigeria	2400
GDP PPP (2,000 to 3,999 $)	Swaziland	1200
	Senegal	1200
	Tanzania	2386
	Guinea	1200
	Ivory Coast	1199
	Zimbabwe	2400
	Kenya	2397
	Cameroon	1182
	Benin	1200
	Lesotho	1200
	Mali	1200
	Sudan	1200
GDP PPP (1,500 to 1,999 $)	Burkina Faso	1200
	Togo	1200
	Sierra Leon	1191
	Uganda	2400
	Madagascar	1200
GDP PPP (Less than 1,500$)	Niger	1200
	Mozambique	2400
	Malawi	2400
	São Tomé and Principe	1196
	Liberia	1199
	Burundi	1200

***Reference**: Micro data analysis of Afrobarometer's data in 2016; World Bank, 2018.*

Figure 2: This is Ahmadi's (2019) Table 1 from the original study.

Table 2. Characteristics of sample of SSA countries (n= 49,137).

		n=	%
Area of residence	Rural	29,312	60.4
	Urban	19,225	39.6
Sex	Female	24,736	50.3
	Male	24,401	49.7
Age groups	18-25	11944	24.5
	26-35	14876	30.5
	36-45	9885	20.2
	46-55	6023	12.3
	56-65	3708	7.6
	66 and over	2414	4.9
Education	No formal education	9223	18.8
	Primary	14403	29.4
	Secondary	18323	37.4
	Post-secondary	7057	14.4
Employment status	Unemployed	30045	61.4
	Employed part-time	5898	12.1
	Employed full-time	12966	26.5
Income availability	Always	7416	15.1
	Many times	13028	26.5
	Several times	11485	23.4
	Just once or twice	5823	11.9
	Never	11176	22.7
Asset ownership (Accumulated score)	0	5544	11.3
	1	10073	20.5
	2	12949	26.4
	3	13896	28.3
	4	6654	13.5
Electricity	No	18662	38.0
	Yes	30459	62.0
Water insecurity	Always	4127	8.4
	Many times	5592	11.4
	Several times	7872	16.1
	Just once or twice	5523	11.3
	Never	25929	52.9
Food insecurity	Always	1290	2.6
	Many times	5552	11.3
	Several times	9313	19.0
	Just once or twice	7249	14.8
	Never	25630	52.3

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Reference: *Micro data analysis of Afrobarometer's data in 2016.*

Figure 3: This is Ahmadi's (2019) Table 2 from the original study.

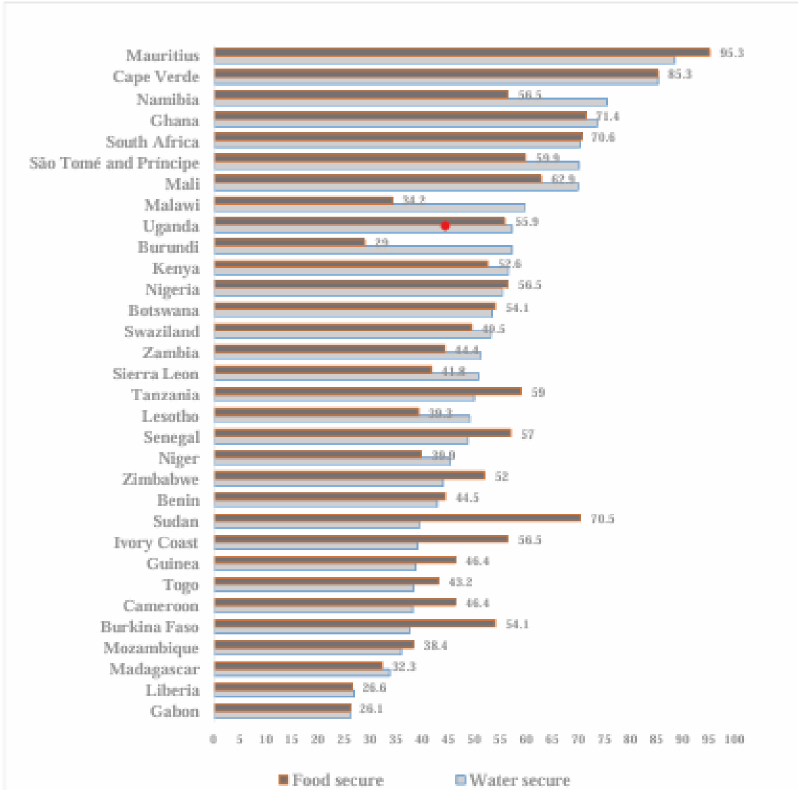

Figure 1. Water secure (nevere gone) and food secure (nevere gone) in SSA 2016 (Reference: Micro data analysis of Afrobarometer's data in 2016.)

Figure 4: This is Ahmadi's (2019) Figure 1 from the original study.

Table 3. Crosstabs tables between food insecurity, water insecurity and socio-economic factors in SSA countries (n=49,137).

	Strength	Sig level
Water insecurity	0.485	0.000
Gender	0.027	0.000
Age group	0.071	0.000
Area of residence	0.137	0.000
Education	-0.283	0.000
Employment	-0.233	0.000
Income availability	0.581	0.000
Asset ownership	-0.334	0.000

Reference: *Micro data analysis of Afrobarometer's open data in 2016*

Figure 5: This is Ahmadi's (2019) Table 3 from the ofiginal study.

Table 4. Unadjusted multinomial logistic regression analyses between Food Insecurity and Water Insecurity by controlling socio-demographic factors in sub-Saharan Africa (n=49,137)

		Never	Just once or twice a month	Several times	Many times
		95% CI for EXP (B)	95% CI for EXP (B)	95% CI for EXP (B)	95% CI for EXP (B)
Water insecurity	Never	39.17 (33.40-45.93)	12.99 (10.91-15.47)	7.90 (6.70-9.31)	4.609 (3.89-5.45)
	Just once or twice	16.57 (12.82-21.40)	27.82 (21.63-36.25)	10.50 (8.09-13.62)	3.609 (2.74-474)
	Several times	11.24 (9.16-13.79)	13.79 (11.11-17.13)	13.46 (10.95-16.54)	5.424 (4.38-6.71)
	Many times	3.494 (2.949-4.13)	3.62 (3.00-4.37)	4.023 (3.38-4.77)	4.520 (1.06-5.72)
	Always (Ref)				
Gender	Female	0.828 (0.740-0.926)	0.891 (0.791-1.003)	0.880 (0.783-0.989)	0.956 (0.846-1.079)
	Male (Ref)				
Age groups	18-25	2.191 (1.716-2.798)	2.266 (1.737-2.956)	1.515 (1.174-1.956)	1.328 (1.017-1.734)
	26-35	1.716 (1.358-2.167)	1.942 (1.504-2.507)	1.392 (1.091-1.777)	1.246 (0.956-1.608)
	36-45	1.400 (1.101-1.781)	1.583 (1.217-2.059)	1.276 (0.992-1.640)	1.202 (0.924-1.563)
	46-55	1.533 (1.179-1.992)	1.599 (1.202-2.127)	1.313 (0.999-1.726)	1.450 (1.091-1.927)
	56-65	1.328 (1.000-1.796)	1.451 (1.066-1.974)	1.137 (0.846-1.529)	1.430 (1.053-1.943)
	66 and over (Ref)				
Area of residence	Rural	0.457 (0.403-0.518)	0.630 (0.552-0.719)	0.756 (0.663-0.861)	0.984 (0.858-1.127)
	Urban (Ref)				
Employment	Unemployed	0.303 (0.257-0.357)	0.438 (0.370-0.520)	0.636 (0.537-0.753)	0.762 (0.639-0.908)
	Employed full-time	0.538 (0.424-0.682)	0.880 (0.688-1.125)	0.903 (0.706-1.154)	0.653 (0.739-1.231)
	Employed part-time (Ref)				
Income availability	Always	83.95 (63.89-110.3)	8.24 (6.18-11.00)	2.76 (2.07-3.70)	1.218 (0.890-1.668)
	Many times	44.2 (31.9-	34.7 (24.9-48.3)	4.83 (3.45-677)	1.724 (1.201-

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		Never	Just once or twice a month	Several times	Many times
		61.31)			2.474)
	Several times	23.8 (19.4-29.1)	16.5 (13.4-20.3)	10.2 (8.38-12.58)	2.446 (1.974-3.032)
	Just once or twice	9.22 (7.88-10.7)	7.55 (6.40-8.9)	6.88 (5.88-.04)	5.350 (4.568-6.266)
	Never (Ref)				
Asset ownership	0	0.065 (0.050-0.085)	0.208 (0.157-0.276)	0.398 (0.301-0.526)	0.873 (0.651-1.170)
	1	0.168 (0.129-0.220)	0.425 (0.322-0.562)	0.729 (0.552-0.963)	1.171 (0.873-1.5690
	2	0.270 (0.207-0.354)	0.677 (0.512-0.896)	0.927 (0.701-1.122)	1.113 (0.828-1.496)
	3	0.490 (0.372-0.645)	0.807 (0.606-1.074)	0.895 (0.672-1.192)	0.921 (0.680-1.249)
	4 (Ref)				
Education	No formal education	0.141 (0.111-0.179)	0.250 (0.194-0.322)	0.510 (0.397-654)	0.975 (0.749-1.269)
	Primary	0.238 (0.187-0.302)	0.515 (0.402-0.660)	0.722 (0.563-0.925)	1.113 (0.856-1.447)
	Secondary	0.536 (0.420-0.685)	0.847 (0.657-1.092)	0.985 (0.764-1.270)	1.083 (0.827-1.419)
	Post-secondary (Ref)				

Reference: Micro data analysis of Afrobarometer's data in 2016

Figure 6: This Ahmadi's (2019)Table 4 from the original study.

Table 5. Adjusted multinomial logistic regression analyses between water insecurity and food insecurity by controlling factors[1] in sub-Saharan Africa (n=49,137)

		Never	Just once or twice a month	Several times	Many times
		95% CI for EXP (B)	95% CI for EXP (B)	95% CI for EXP (B)	95% CI for EXP (B)
Water insecurity	Never	34.08 (28.96-40.11)	12.06 (10.09-14.41)	7.648 (6.467-9.043)	4.555 (3.836-5.409)
	Just once or twice	14.90 (11.47-19.34)	26.05 (19.90-34.9)	10.275 (7.886-13.8)	3.509 (2.654-4.638)
	Several times	10.51 (8.51-12.98)	13.41 (10.73-16.76)	13.60 (11.00-16.81)	5.611 (4.50-6.981)
	Many times	3.351 (2.81-3.994)	3.625 (2.987-4.399)	4.093 (3.430-4.884)	5.024 (4.210-5.996)
	Always (Ref)				
Gender	Female	0.845 (0.749-0.953)	0.891 (0.785-1.011)	0.874 (0.772-0.989)	0.961 (0.846-1.092)
	Male (Ref)				
Age groups	18-25	2.161 (1.660-2.814)	2.202 (1.660-2.920)	1.521 (1.161-1.992)	1.343 (1.016-1.776)
	26-35	1.501 (1.164-1.936)	1.721 (1.310-2.262)	1.346 (1.037-1.746)	1.225 (0.936-1.603)
	36-45	1.168 (0.899-1.517)	1.359 (1.026-1.799)	1.223 (0.936-1.598)	1.180 (0.895-1.555)
	46-55	1.375 (1.035-1.726)	1.484 (1.095-2.011)	1.347 (1.007-1.800)	1.511 (1.112-2.036)
	56-65	1.209 (0.892-1.369)	1.360 (0.983-1.883)	1.130 (0.828-1.542)	1.434 (1.043-1.972)
	66 and over (Ref)				
Area of residence	Rural	0.697 (0.610-0.796)	0.886 (0.771-1.018)	0.983 (0.858-1.127)	1.190 (1.033-1.371)
	Urban (Ref)				
Employment	Unemployed	0.376 (0.316-0.446)	0.502 (0.419-0.600)	0.783 (0.618-0.881)	0.835 (0.695-1.003)
	Employed full-time	0.627 (0.490-0.802)	0.915 (0.710-1.179)	0.959 (0.745-1.234)	1.008 (0.777-1.309)
	Employed part-time (Ref)				

Model fitting Criteria -2 Log Likelihood=6362.673; Chi-sq=11520.316; df=52; Sig=0.000
Pseudo R-Square= Cox and Snell= 0.214; Nagelkerke= 0.232; McFadden= 0.094
***Reference**: Micro data analysis of Afrobarometer's data in 2016*

Figure 7: This Ahmadi's (2019) Table 5 from the original study.